**King Fahd University of Petroleum and Minerals**
**Department of Mathematics**
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# Math 201 Major Exam I Term 233

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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**1.** The graph of the parametric equations

$$x = \cos t, \quad y = 2\cos t + 1$$

is

(a) a hyperbola

(b) a circle

(c) an ellipse

(d) a parabola

(e) a line segment

**2.** The area of the region outside the circle $r = 2$ and inside the circle $r = 4\sin\theta$ is equal to

(a) $\dfrac{4\pi}{3} + 2\sqrt{3}$

(b) $\dfrac{5\pi}{3} - 2\sqrt{3}$

(c) $\dfrac{2\pi}{3} + 2\sqrt{3}$

(d) $\dfrac{4\pi}{3} + \sqrt{3}$

(e) $\dfrac{2\pi}{3} + \sqrt{3}$

**3.** The slope of the tangent line to the parametric curve

$$x = t^2 - t, \quad y = t^3 - 3t - 1$$

at the point corresponding to $t = 2$ is equal to

(a) $-\dfrac{1}{3}$

(b) 4

(c) 6

(d) 3

(e) $-2$

**4.** If $C(a, b, c)$ is the center of the sphere

$$3x^2 + 3y^2 + 3z^2 - 18x + 6z + 24 = 0,$$

then $a^2 + b^2 + c^2 =$

(a) 11

(b) 3

(c) 10

(d) 5

(e) 14

**5.** The vector of magnitude $\frac{3}{2}$ that has the same direction as the vector $\langle 2, -2, 1\rangle$ is

(a) $\left\langle -3, 3, \frac{3}{2} \right\rangle$

(b) $\left\langle 1, -1, \frac{1}{2} \right\rangle$

(c) $\left\langle -1, \frac{1}{2}, 1 \right\rangle$

(d) $\langle 2, -1, 2\rangle$

(e) $\langle 2, -2, 1\rangle$

**6.** The arc length of the parametric curve

$$x = e^t \cos t, \quad y = e^t \sin t, \quad 0 \le t \le 1$$

is equal to

(a) $e + 1$

(b) $2(e - 1)$

(c) $\sqrt{2}\,(e + 1)$

(d) $\sqrt{2}\,(e - 1)$

(e) $\sqrt{2}\,(e^2 - 1)$

**7.** The projection of $\vec{u} = \langle 1, -2, 3\rangle$ onto $\vec{v} = \langle -1, 1, 2\rangle$ is

(a) $\left\langle -\frac{1}{2}, \frac{1}{2}, 1 \right\rangle$

(b) $\langle 2, -2, -4\rangle$

(c) $\langle -1, 1, 2\rangle$

(d) $\left\langle \frac{1}{2}, -1, \frac{3}{2} \right\rangle$

(e) $\langle -2, 2, 4\rangle$

**8.** The area of two petals (two loops) of the rose curve $r = \sin(8\theta)$ is equal to

(a) $\dfrac{\pi}{4}$

(b) $\dfrac{5\pi}{16}$

(c) $\dfrac{\pi}{16}$

(d) $\dfrac{3\pi}{8}$

(e) $\dfrac{\pi}{8}$

**9.** Converting the rectangular equation

$$(x^2 + y^2)^2 - 4x^2 + 4y^2 = 0$$

to polar form, we get

(a) $r^2 = 4\cos(2\theta)$

(b) $r = 2\sin(2\theta)$

(c) $r^2 = 4\sin(2\theta)$

(d) $r^2 = 2\cos(2\theta)$

(e) $r^2 = 4(\cos\theta - \sin\theta)$

**10.** The polar curve $r = 2\csc\theta + 3$ has a horizontal tangent line when

(a) $\theta = \dfrac{\pi}{8}$

(b) $\theta = \dfrac{\pi}{3}$

(c) $\theta = \dfrac{\pi}{6}$

(d) $\theta = \dfrac{\pi}{4}$

(e) $\theta = \dfrac{\pi}{2}$

**11.** The curve represented by the parametric equations

$$x = \sec^2 t, \quad y = \cos t, \quad 0 \le t < \frac{\pi}{2}$$

is represented by the rectangular equation

(a) $y = -\dfrac{1}{\sqrt{x}},\ x \ge 1$

(b) $y = \dfrac{1}{x^2},\ x > 0$

(c) $y = \dfrac{1}{\sqrt{x}},\ x \ge 1$

(d) $y = \dfrac{1}{\sqrt{x}},\ x > 0$

(e) $y = \dfrac{1}{x^2},\ x \ge 1$

**12.** The area of the surface formed by revolving the circle $r = \cos\theta$ about the line $\theta = \dfrac{\pi}{2}$ is given by

(a) $\int_0^{\pi} \pi \cdot \sin\theta \cdot \cos\theta \, d\theta$

(b) $\int_0^{2\pi} 2\pi \cdot \sin\theta \cdot \cos\theta \, d\theta$

(c) $\int_0^{\pi} 2\pi \cdot \cos^2\theta \, d\theta$

(d) $\int_0^{\pi} \pi \cdot \cos^2\theta \, d\theta$

(e) $\int_0^{2\pi} 2\pi \cdot \cos^2\theta \, d\theta$

**13.** A vector that is orthogonal to both vectors

$$\vec{u} = \mathbf{i} + \mathbf{j} + \mathbf{k}, \quad \vec{v} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$$

is

(a) $3\mathbf{j} - \mathbf{k}$

(b) $-2\mathbf{i} - \mathbf{j} + \mathbf{k}$

(c) $2\mathbf{i} + 3\mathbf{j}$

(d) $-2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$

(e) $-2\mathbf{i} - 3\mathbf{j} - \mathbf{k}$

**14.** The volume of the parallelepiped having adjacent edges

$$\vec{u} = \langle 3, 1, 1 \rangle, \quad \vec{v} = \langle 2, 1, 0 \rangle, \quad \vec{w} = \langle 1, -1, -2 \rangle$$

is equal to

(a) 6

(b) 10

(c) 2

(d) 8

(e) 5

**15.** If $R(a, b, c)$ is the point that lies three-fifths of the way from $P(0, 1, 2)$ to $Q(2, -1, -3)$, then $a + b + c =$

(a) $-\frac{1}{5}$

(b) 1

(c) $\frac{3}{5}$

(d) $-3$

(e) 0

# Answer Key

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| Q | Ans | Brief explanation |
|---|---|---|
| 1 | **(e)** | $y = 2x + 1$ with $-1 \le x \le 1$: a line segment. |
| 2 | **(a)** | Limits $\pi/6$ to $\pi/2$ (doubled): $\int (4 - 8\cos 2\theta) d\theta = \frac{4\pi}{3} + 2\sqrt{3}$. |
| 3 | **(d)** | $\frac{dy}{dx} = \frac{3t^2-3}{2t-1}$, at $t = 2$ equals $\frac{9}{3} = 3$. |
| 4 | **(c)** | $(x-3)^2 + y^2 + (z+1)^2 = 2$, center $(3, 0, -1)$, so $9 + 0 + 1 = 10$. |
| 5 | **(b)** | $\frac{3}{2} \cdot \frac{1}{3}\langle 2, -2, 1\rangle = \langle 1, -1, \frac{1}{2}\rangle$. |
| 6 | **(d)** | $(x')^2 + (y')^2 = 2e^{2t}$, $L = \int_0^1 \sqrt{2}e^t \, dt = \sqrt{2}(e-1)$. |
| 7 | **(a)** | $\frac{3}{6}\langle -1, 1, 2\rangle = \langle -\frac{1}{2}, \frac{1}{2}, 1\rangle$. |
| 8 | **(c)** | One petal $= \frac{1}{2}\int_0^{\pi/8} \sin^2 8\theta \, d\theta = \frac{\pi}{32}$, two petals $= \frac{\pi}{16}$. |
| 9 | **(a)** | $r^4 = 4r^2(\cos^2\theta - \sin^2\theta)$, so $r^2 = 4\cos 2\theta$. |
| 10 | **(e)** | $y = 2 + 3\sin\theta$, $dy/d\theta = 3\cos\theta = 0$ at $\theta = \pi/2$ ($dx/d\theta = -5$). |
| 11 | **(c)** | $x = 1/y^2$, $y > 0$, $x = \sec^2 t \ge 1$: $y = 1/\sqrt{x}$, $x \ge 1$. |
| 12 | **(c)** | $S = \int_0^\pi 2\pi \, r\cos\theta\sqrt{r^2 + r'^2} \, d\theta = \int_0^\pi 2\pi\cos^2\theta \, d\theta$. |
| 13 | **(d)** | $\vec{u} \times \vec{v} = -2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$. |
| 14 | **(e)** | $\det = -5$, so $V = \lvert -5\rvert = 5$. |
| 15 | **(e)** | $R = P + \frac{3}{5}\overrightarrow{PQ} = (\frac{6}{5}, -\frac{1}{5}, -1)$, so $a + b + c = 0$. |